

MA1513 Exam
AY2022/2023 Semester I

15 MCQ/FIB to be answered in Exemplify; 4 written questions (LQ) to be written by hand and submitted to Canvas.

Question #: 1

$$\left(\begin{array}{ccccc|c} 1 & 2 & 1 & 0 & 2 & a \\ 0 & 0 & 1 & 0 & 1 & b \\ 0 & 0 & 2 & 0 & 2 & c \end{array} \right)$$

The above is an augmented matrix of a linear system with a, b, c being the constant terms. Which of the following statements are true? (Pick all correct options)

- A. This system is consistent only when $c = 2b$.
- B. If the system is consistent, the general solution has two parameters.
- C. For some values of a, b, c , the system has exactly one solution.
- D. The last column (constant terms) of a row echelon form of the augmented matrix can be a pivot column.
- E. If $c = 0$, the system will have infinitely many solutions.

Item Weight: 3.0

Question #: 2

Given that $\mathbf{A}$ is a singular $n \times n$ matrix, which of the following statements are true? (Pick all correct options)

- A. The rank of $\mathbf{A}$ is n .
- B. The determinant of $\mathbf{A}$ is 0.
- C. The homogeneous system $\mathbf{Ax} = \mathbf{0}$ does not have a trivial solution.
- D. The n column vectors of $\mathbf{A}$ form a linearly dependent set.
- E. There is a non-zero vector in the nullspace of $\mathbf{A}$.

Item Weight: 3.0

Question #: 3

$$\begin{vmatrix} a & b & c \\ a & 0 & c \\ * & * & * \end{vmatrix} = 0$$

The left-hand side above is a 3 x 3 determinant, where a, b, c are non-zero distinct numbers. Which of the following can be the third row of the determinant, regardless of the values of a, b, c? (Pick all correct options)

- A. (a b c)
- B. (0 p 0) where p is a non-zero number.
- C. (a n c) where n is not equal to 0 and b.
- D. (a a a)
- E. (c b a)

Item Weight: 3.0

Question #: 4

Which of the following linear spans do not contain the vector (1,2,3,4) ? (Pick all correct answers)

- A. Span{ (4,3,2,1) }
- B. Span{ (1,2), (3,4) }
- C. Span{ (1,0,3,0), (0,2,0,4) }
- D. Span{ (2,3,4,1), (3,4,1,2), (4,1,2,3) }
- E. Span{ (4,0,0,0), (0,3,0,0), (0,0,2,0), (0,0,0,1) }

Item Weight: 3.0

Question #: 5

Which of the following sets of vectors in $\mathbf{R}^3$ all lie on the same plane? (Pick all correct answers)

- A. (0, 0, 0), (1, 2, 0), (2, 0, 1)
- B. (1, 2, 1), (2, 1, 1), (1, 1, 2)
- C. (1, 2, 3), (4, 5, 6), (7, 8, 9)
- D. (2, 2, 0), (0, 0, 3), (1, 1, 1), (3, 3, 2)
- E. (1, 0, 1), (0, 1, 1), (0, 0, 1)

Item Weight: 3.0

Question #: 6

The projection of the vector $(1,2,3,0)$ onto the subspace $V = \text{span}\{ (1,0,1,0), (0,1,0,1) \}$ is given by $\mathbf{p} = (a, b, c, d)$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$

1.
2.
3.
4.

Item Weight: 3.0

Question #: 7

Let $\mathbf{v}$ be a vector in $\mathbf{R}^3$ and $S = \{ (1,1,0), (1,0,1), (0,1,1) \}$ and $T = \{ (3,0,0), (0,4,0), (0,0,5) \}$ be two bases for $\mathbf{R}^3$. Suppose the coordinate vector of $\mathbf{v}$ with respect to S is $[\mathbf{v}]_S = (1,2,3)$. The coordinate vector of $\mathbf{v}$ with respect to T is $[\mathbf{v}]_T = (a, b, c)$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$

1.
2.
3.

Item Weight: 3.0

Question #: 8

Let $\mathbf{A}$ be a 3×5 matrix. Which of the following statements are true? (Pick all correct options)

- A. The row space of $\mathbf{A}$ is a subspace of $\mathbf{R}^3$.
- B. The sum of two columns of $\mathbf{A}$ is again a column of $\mathbf{A}$.
- C. $\mathbf{A}$ cannot be a full rank matrix.
- D. The nullity of $\mathbf{A}$ is at least 2.
- E. The nullspace of $\mathbf{A}$ is a subspace of $\mathbf{R}^5$.

Item Weight: 3.0

Question #: 9

Suppose $T: \mathbf{R}^2 \rightarrow \mathbf{R}^2$ is a linear transformation and we are given

$$T\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ and } T\left(\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

The standard matrix of T is given by $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$.

1.
2.
3.
4.

Item Weight: 3.0

Question #: 10

$$\mathbf{A} = \begin{pmatrix} 2 & * & * \\ 0 & 3 & * \\ 0 & 0 & 4 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & * & * \\ 0 & 2 & * \\ 0 & 0 & 3 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 2 & * & * \\ 0 & 2 & * \\ 0 & 0 & 2 \end{pmatrix}$$

Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ be the above three upper triangular matrices where each $*$ can represent any number. Which of the following statements are true? (Pick all correct answers)

- A. Matrix $\mathbf{A}$ may or may not be diagonalizable.
- B. Matrix $\mathbf{B}$ may or may not be diagonalizable.
- C. Matrix $\mathbf{C}$ must be non-diagonalizable.
- D. All the eigenspaces of matrix $\mathbf{A}$ must have dimension 1.
- E. Matrix $\mathbf{B}$ must have an eigenspace with dimension 2.

Item Weight: 3.0

Question #: 11

Let $\mathbf{P}$ be a non-singular 3×3 matrix with columns $\mathbf{C}_1$, $\mathbf{C}_2$ and $\mathbf{C}_3$ in that order. Suppose

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which of the following statements are true?

- A. The nullspace of $\mathbf{A}$ is $\text{span}\{\mathbf{C}_2\}$.
- B. $\mathbf{C}_1 + \mathbf{C}_2$ is an eigenvector of $\mathbf{A}$.
- C. $\mathbf{A}^2 = \mathbf{A}$
- D. The matrix with columns $3\mathbf{C}_1$, $-\mathbf{C}_2$, $2\mathbf{C}_3 + \mathbf{C}_1$ will diagonalize matrix $\mathbf{A}$.
- E. $\mathbf{A}$ is a non-singular matrix.

Item Weight: 3.0

Question #: 12

A certain 3rd order ODE is given by $y''' + ay'' + by' + cy = 0$. This ODE can be converted to a 3×3 SDE given by

$$\mathbf{y}' = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -1 & 3 \end{pmatrix} \mathbf{y}.$$

Then $a = \underline{\quad 1 \quad}$, $b = \underline{\quad 2 \quad}$, $c = \underline{\quad 3 \quad}$.

- 1. _____
- 2. _____
- 3. _____

Item Weight: 3.0

Question #: 13

An SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ gives an equilibrium point which is a degenerate node. Suppose the matrix $\mathbf{A}$ has an eigenvalue 3 with an eigenvector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and corresponding generalised eigenvectors $\begin{pmatrix} 1/3 \\ s \end{pmatrix}$ for any s . Which of the following are solutions of the SDE? (Pick all correct answers)

A. $\begin{pmatrix} 0 \\ 3 \end{pmatrix} e^{3t}$

B. $\begin{pmatrix} 1 \\ t \end{pmatrix} e^{3t}$

C. $\begin{pmatrix} 1/3 \\ 0 \end{pmatrix} e^{3t}$

D. $\begin{pmatrix} 1/3 \\ t+1 \end{pmatrix} e^{3t}$

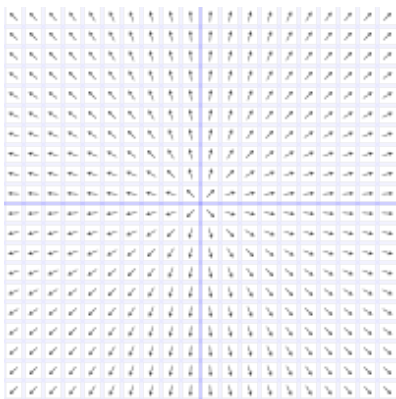
E. $\begin{pmatrix} 1/3 \\ 1 \end{pmatrix} e^{3t}$

Item Weight: 3.0

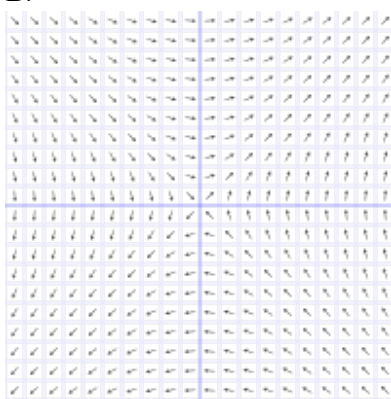
Question #: 14

Which of the following direction fields belong to the SDE $y' = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} y$?

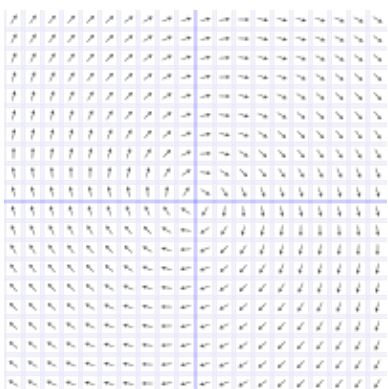
A.



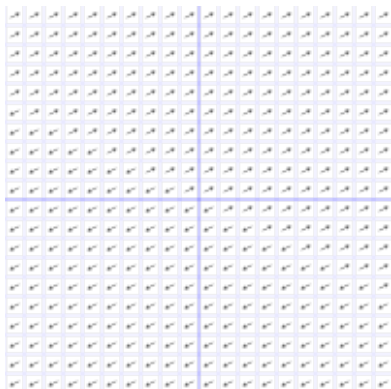
B.



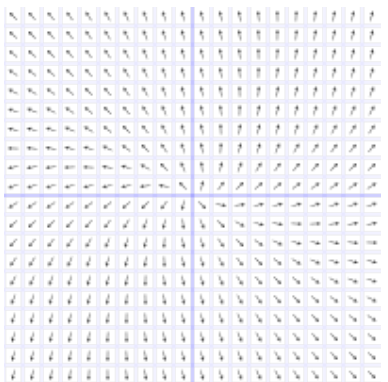
C.



D.



E.



Question #: 15

Suppose a 2 x 2 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ has the following fundamental set of solutions:

$$x_1 = e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cos 3t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \sin 3t, x_2 = e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin 3t - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos 3t$$

Which of the following statements are true? (Pick all correct answers)

- A. The equilibrium point of the SDE is unstable.
- B. The matrix $\mathbf{A}$ has eigenvalues $1+3i$ and $1-3i$.
- C. The matrix $\mathbf{A}$ has an eigenvector $\begin{pmatrix} 1 \\ i \end{pmatrix}$.
- D. The equilibrium point is a center.
- E. The trajectories of the SDE are counter-clockwise.

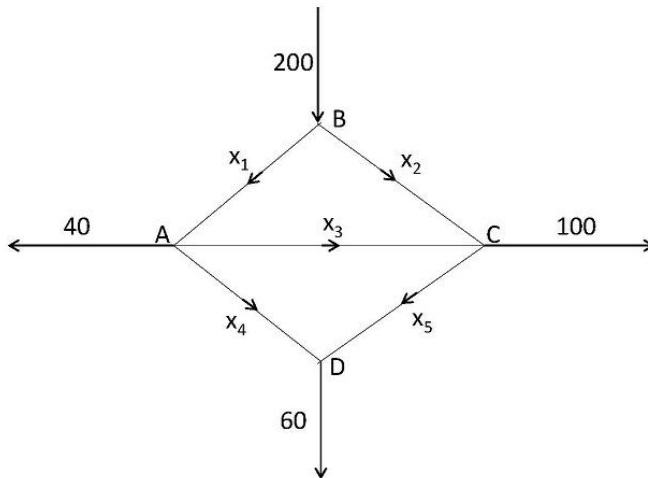
Item Weight: 3.0

Question #: 16

Long Question 1 (LQ1)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ1 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

The network in the diagram shows the traffic flow (in vehicles per hour) over several one-way streets in the city area during peak hours on a typical weekday.



- (a) [3 marks] Set up a linear system $\mathbf{Ax} = \mathbf{b}$ with 4 equations (in the order of the junctions A, B, C and D) and traffic volumes arranged in the order of x_1, x_2, x_3, x_4 and x_5 . Write down the matrix $\mathbf{A}$ and vector $\mathbf{b}$ explicitly.
- (b) [3 marks] Write down the general solution of the linear system in (a). (You may use MATLAB for the working.)
- (c) [2 marks] What is the largest possible value for x_3 , given that all the x_i must be non-negative?
- (d) [2 marks] If the street between A and D is closed, what is the range for the possible values of x_1 ?

Item Weight: 10.0

Question #: 17

Long Question 2 (LQ2)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ2 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

The revenue y (in \$100. i.e. when $y=1$, it means the revenue is \$100) from the sale of x (in 100 of units, i.e, when $x=1$, it means 100 units are sold) of a product by company Z can be described using the equation:

$$y = mx + nx^2 \quad (*)$$

Some data points (x,y) were collected : (1, 1.8), (2, 2.7), (3, 3.4), (4, 3.8), (5, 3.9).

- (a) [3 marks] Set up a linear system $\mathbf{Ax}=\mathbf{b}$ with five equations to find the coefficients a and b in the equation (*). Write down the matrix $\mathbf{A}$ and vector $\mathbf{b}$ explicitly.
- (b) [3 marks] Find the values of m and n that produce the least squares fit of the given data points. (You may use MATLAB for the working. Give your answers up to 4 decimal places.)
- (c) [2 marks] What is the projected revenue when 500 units of the product are sold based on the least squares fit model in (b)? (Give your answer to the nearest dollar.)
- (d) [3 marks] According to the least squares fit model in (b), should company Z sell more than 500 units of the product to increase the revenue? Why?

Item Weight: 11.0

Question #: 18

Long Question 3 (LQ3)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ3 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

In a certain city, a car rental company has 3 picking up and returning locations: airport, train terminal and downtown. The weekly pattern of pick-ups and returns is given below:

- i. 90% of the car picked up from the airport are returned to the airport, while 9% and 1% are returned to the train terminal and downtown respectively;
- ii. 90% of the car picked up from the train terminal are returned to the train terminal, while 9% and 1% are returned to the airport and downtown respectively;
- iii. 90% of the car picked up from downtown are returned to downtown, while 9% and 1% are returned to the airport and train terminal respectively.

(a) [3 marks] Find the 3 x 3 matrix $\mathbf{A}$ such that the matrix equation $\mathbf{y}_n = \mathbf{A}\mathbf{y}_{n-1}$ where

$$\mathbf{y}_n = \begin{pmatrix} a_n \\ t_n \\ d_n \end{pmatrix}$$

represent the number of cars a_n , t_n , d_n ready to be rented out at the three locations: airport, train terminal and downtown, at the beginning of week n .

(b) [3 marks] Find a matrix $\mathbf{P}$ that diagonalizes the matrix $\mathbf{A}$ in part (a). (You may use MATLAB for the working. Give the entries of $\mathbf{P}$ in fractions.)

(c) [3 marks] Use (b) to find $\mathbf{A}^n$ when n tends to infinity. (Give your answer in fractions.)

(d) [3 marks] At the steady state, what is the ratio $a_n : t_n : d_n$ of the cars ready to be rented at the 3 locations?

Item Weight: 12.0

Question #: 19

Long Question 4 (LQ4)

Answer this question by writing in the exam answer booklet. Do not write in the box below. Indicate LQ4 at the top left corner on each page of the answer to this question. You may use MATLAB for this question. But you should show how your answers are derived.

$$\begin{pmatrix} v_1' \\ v_2' \end{pmatrix} = \begin{pmatrix} 4 & -5 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

The above SDE gives a direction field that describes the force field (on the phase plane). Any particle that is placed on the plane will trace out a trajectory according to the force field.

- (a) [3 marks] Sketch the phase portrait of the SDE, indicating the axes, if any, and the directions of the trajectories.
- (b) [3 marks] Write down the general solution of the SDE.
- (c) [3 marks] Suppose the particle is at the position (2.9, 2.6) on the plane, write down the particular solution of the SDE that corresponds to the trajectory of this particle under the force field. (Give your answers in fractions.)
- (d) [3 marks] Is there any position on the plane that we can place a particle so that it will move towards the origin? Justify your answers.

Item Weight: 12.0